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# Section 1: Drafting a Technical Report

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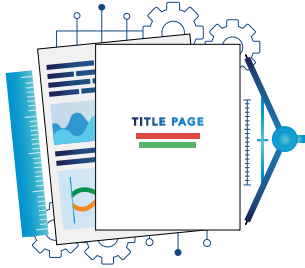


Figure 1 Drafting your technical report Credit: © Course Author(s) and University of Waterloo.

Now you are ready to focus on drafting the report. Writing is hard work, but this class has been organized to get you to this stage so that you can write confidently, and so that you have everything you need to write a terrific report.

## Analysis

The main content of your technical report should be description of the technical processes you have engaged in to address the problem in your workplace.

## Science

For scientists, this can mean describing experimentation, or simply walking your reader through a workplace process, with the goal of learning more about your problem by engaging in the scientific method.

## Math

For mathematicians, this often means describing your analysis – the ways that you use math to try and explore your problem.

## Engineering

For engineers, this will vary according to your workplace problem.

The key is to carefully walk your reader through the technical process: the experimentation or the analysis. Give them a tour of your workplace and detail the steps in your thinking.

## Examples of Technical Processes in a Report

Have a look at the following sentences, pulled from a strong technical report, that puts technical processes in simple terms:

"The "f-words" test requires the patient to say as many words starting with "f" as possible in one minute. The "f-words" test is a measure of verbal fluency and less than 12 words is indicative of impaired verbal fluency."

Other technical processes can't really be simplified, but they can be broken up so that readers understand what is happening, and when. Have a look at this lengthy section:

"Currently, operators are banging 50-lb coconut bags at a rate of 14 bags per hour to manually break the coconut into smaller pieces and dumping them through the hole. The screen size on the hole is a 5" by 5" screen and hence the biggest chunk of coconut that can go through would be a 5" x 5" chunk. The 50-lb bags are 35" x 12" x 4". The hole is attached to an inclined auger that brings the premix materials up the premix bin (shown in Figure 2). So it is a process of ripping the coconut bags open, smashing the bags into smaller pieces so that each chunk of coconut can fit through the hole and dumping the contents of the bag inside the hole. However, doing so on an 8-hour shift every day is straining for the operators as proven by the constant influx of new premix operators for the Harvest line. Due to their back and shoulder pain, operators running the premix line have been changing on a monthly basis.

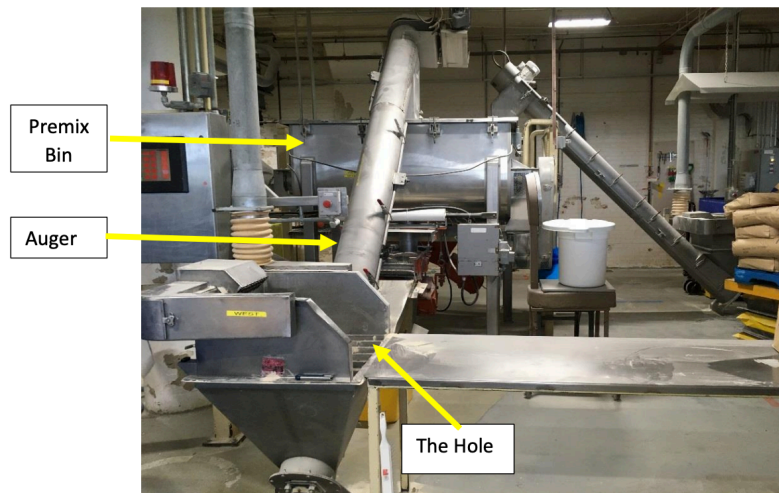


Figure 2: Premix Line

Note. An image depicting the premix section of the harvest line, which uses gravity to direct the premix materials to the production line. Image captured by author, 2020."

As seen above, because the process is particularly complicated, the author has also created a Figure (Figure 2) to illustrate the process. The legend of the figure is sufficiently detailed so that the reader understands the data presented.

## Waterloo Writing Works: Figures and Tables

Tables present lists of numbers or text in columns, each column having a title or label. Figures, on the other hand, are visual presentations of results, including graphs, diagrams, photos, drawings, schematics, maps, etc.

For your interest, select the button to learn more about figures and tables.

[Figures and Tables ↗](#)

It's your job to break things up as much as possible, simplify your writing as much as you reasonably can, and walk your audience patiently through your processes and analyses.

At this stage, you want to be sure you feel confident with some of the basics of writing your report, including common punctuation. It may seem overly simple, but punctuation is what gives your report a rhythm: punctuation is how we introduces pauses, stops, and even extensions into our sentences and paragraphs.

## Waterloo Writing Works: Common Punctuation

You need to feel confident that you know how to properly use common punctuation, including periods, commas, semi-colons and colons.

For your interest, select the button to learn more about common punctuation.

[Common Punctuation](#) 

## Drafting

Now you need to begin translating your technical processes and analyses into words. What do you need to explain to your audience so that they understand your problem and the steps you have taken to address it?

To begin with, simply speak these technical details and steps into a device - you can record this as an audio file or use speech recognition on a phone or computer to record this as text. Listen or look at what you have and begin to transform this into simple language in a word processor.

## Waterloo Writing Works: Simple Language

Utilize short and simple sentences, using simple words, in active voice, in third person, and organized in chronological order. You will want to speak in present tense as reports are generally written in this tense. The goal is to guide your reader through your process and your thinking so that they can reproduce the exact same results: this should be like a great recipe or an excellent map. **Think:** if I were thrust into this workplace, could I follow this writing and know what to do to replicate these processes or analyses?

For your interest, select the buttons to learn more about how to apply simple words, simple sentences, tense, and agreement to your technical report writing.

[Simple Words ↗](#)[Simple Sentences ↗](#)[Tense ↗](#)[Agreement ↗](#)

## Letter of Submittal

The Letter of Submittal in your report will not be addressed to an evaluator in your faculty. Since it is being submitted to PD11 Evaluators, you can address them as “Dear Evaluators.” The address that you use can be:

Evaluators

WatPD

University of Waterloo

Waterloo, Ontario

N2L 3G1

Otherwise, you do want to include the same of the details that most Letters of Submittal have. You can model your letter on these two examples from award-winning reports shown below. Make sure you fill in as many of these same details as you can. But also, tell your evaluator:

- A little bit about what you have learned from the revision process — what was the most useful feedback you received from peers or experts?
- What major changes did you make?
- What have you learned about reports, about the report-writing process, and about yourself as a writer through this class?

This listing of questions offers an example of parallel structure, one of the ways we can make our writing more readable and more elegant. Review Waterloo Writing Works module on **parallel structure** , and look for ways to utilize this technique in your final draft before you submit it.

## Dynamic Data Storing Solution in Data Management System

(Yiqiao Sun, Mathematics)

### Memorandum



To: Faisal Shaheen

From: Yiqiao Sun

Date: Feb 25, 2020

Re: Technical Report: Dynamic Data Storing in  
Data Management System

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I have enclosed my technical report, "Dynamic Data Storing Solution in Data Management System", written on my 2A work term in the City of Toronto, Toronto Water division. This report has not yet received any academic credit, and the successful accomplishment of which is part of PD11 Process for Technical Report Writing a mandatory component of my Mathematics / Financial Analysis and Risk Management Co-op Program Degree

As an IT trainee in the EWMS (Enterprise Working Management Solution) team, my job is to ensure that Toronto Water's business requirements and customer needs are communicated. My daily duties are assisting in updating and maintaining statistical records on projects, supporting the implementation of a new data system for migration of documents and grouping data files. Based on my learning from my responsibilities, the report is addressed to report and give possible solutions to the updating data storing problem.

Thank you for your assistance in preparing this report.

Sincerely,

Yiqiao Sun

Notice the tense of this paragraph. In almost all cases, the technical report should be written in the present tense. When referring to your job, if you are still in your co-op term, use the present tense in this Letter of Submittal as well. There is a Waterloo Writing Works Module on tense — and the common errors we make with it — that will help you understand when to use present and when to use past tense.

## **Characterization of Chemicals Released in the City of Ottawa's Wastewater and Risk Assessment Strategies**

(Carlene Perkin, Biology)

### **Letter of Submittal**

Carlene Perkin

January 5, 2015

Dr. David R. Rose  
Biology Department  
University of Waterloo  
200 University Avenue West  
Waterloo, Ontario  
N2L 3G1

Dear Dr. Rose:

My report, entitled "Characterization of Chemicals Released in the City of Ottawa's Wastewater and Risk Assessment Strategies" was written on my 2B work term for the City of Ottawa. The intention of this report is to be a literature review of the compounds in the City of Ottawa's treated wastewater effluent, and to discuss deleterious and unknown effects on aquatic life in the receiving environment.

The City of Ottawa's Water Environment Protection Program, in which I was employed, is supervised by Brian Bezaire and strives to assess and enhance the quality of surface water resources. The Program studies fish species, benthos diversity, and water chemistry of the city's rivers and streams on an ongoing basis.

This technical report was created by me and has not received any previous academic credit at this or any other institution. I would like to thank Mr. Brian Bezaire for providing constructive guidance and resources, and for proofreading my report.

I have read and understood my faculty's technical report guidelines before producing my report. I have checked the report for spelling and grammatical mistakes. This report discusses aspects of my co-op project.

Sincerely,

Carlene Perkin

One common error when using "modal verbs" like **have** is that the other main verbs in the sentence (read and understood) don't **agree** with them. There is a Waterloo Writing Works module on this common error and you should make sure to review it and apply its lessons to your final report before you submit it.

Please refer to the  
**Assignment 3: First Draft of  
Technical Report** for full  
assignment details, including  
submission information.